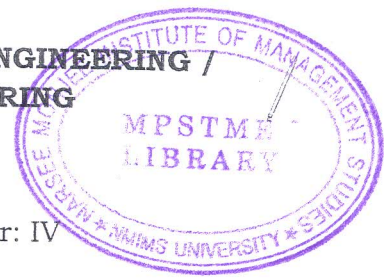


SVKM's NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING /
SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2021-22



Programme: B. Tech (Data Science)

Year: II

Semester: IV

Subject: Introduction to Data, Signal and Image Analysis

Date: 09 June 2022

Marks: 100

Time: 10.00 am to 1.00 pm

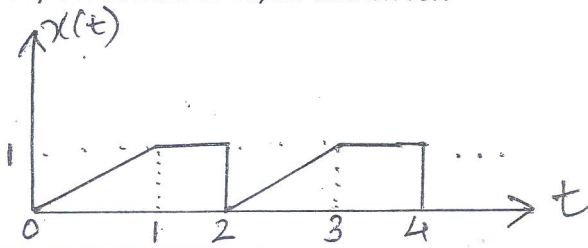
Durations: 3 (Hrs)

No. of Pages: 02

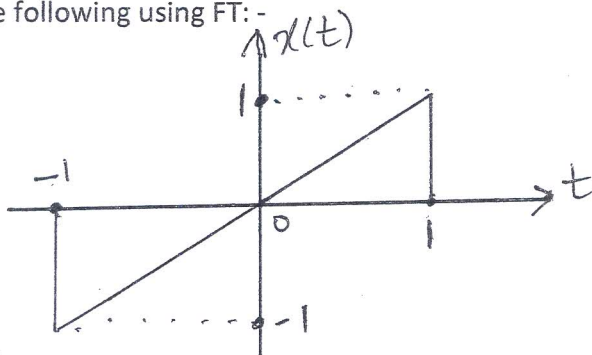
Re-Examination (2021-22)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) In all 5 questions to be attempted.
- 4) All questions carry equal marks.
- 5) Answer to each new question to be started on a fresh page.
- 6) Figures in brackets on the right hand side indicate full marks.
- 7) Assume Suitable data if necessary.

Q1		Answer briefly:	marks
CO1; SO2; BL1	a.	Define even and odd signals	5
CO3; SO1; BL2	b.	Explain Butterworth filter	5
CO1; SO1; BL1	c.	Define ROC for Z-transform	5
CO2; SO6; BL4	d.	Compare fir vs iir	5
Q2 CO3; SO1; BL5	a.	What are the possible outcomes $Y[n]$ for: - $x[n] = 12, 14, 18, 19, 23$ and $h[n] = 3, 4, 6, 11, 12$	10
Q2 CO1; SO6; BL3	b.	Identify the results of $a0$, a_n and b_n for: - 	10

1/2

SO2;	a.	Discuss Sampling in detail	10
SO1;	b.	How would you solve circular convolution for the periodic sequences 1,2,3 and 2,1	10
SO6;	a.	Plot $x(t) = 2u(t+4) - u(t+2) + r(t) - r(t-2) + u(t-4) - r(t-6) + r(t-8)$ and hence how would you improve on the given question (in terms of unit and ramp) to make it more comprehensible	10
SO7;	b.	Write a program to perform enhancement and reduction of an image using Log and power transformations (in any programming language of your choice)	10
SO6;	a.	Interpret the following using FT: - 	10
SO7;	b.	Write a program obtain the output for averaging filter and median filter (in any programming language of your choice)	10
SO1;	a.	How would you outline all the possible ROC for Laplace	10
SO1;	b.	How would you compare Digital and Analog Signals	10